وزارة الإتصالات
وتكنولوجيا المعلومات

Probability and Statistics for Machine Learning

AI TALENTS HUB

EPITA
ÉCOLE D'INGÉNIEURS EN INFORMATIQUE

AI-Pro Program Structure

Foundation Period (3 Month)

- Knowledge Representation & AI History
- Introduction To Python
- Probability For Machine Learning
- Linear Algebra For Data Science
- Numerical Algorithms
- Optimization For Data Science
- Java & UML Programming
- Data Exploration & Preparation
- Algorithm Workshop: Initiation
- Java Programming Basics (Online)
- Linux Administration

Excellence Period (3 Month)

- Intermediate Python for Data Science
- NOSQL Databases
- Machine Learning I: Introduction & Statistical ML
- Machine Learning II: Bayesian & Unsupervised Methods
- Neural Networks & Deep Learning
- Reinforcement Learning
- Spark & Python for Big Data
- Big Data Infrastructure & Cloud Computing
- Data Visualization

Operational Period (3 Month)

AI in Image and Video Processing

- AI in Signal & Audio Processing
- Deep Learning for Natural Language Processing
- Recommender System
- Data Science in Production
- Ethical Development of AI Applications
- AI Project Methodology & Industrialization
- Action Learning: Various Domains of Applications

Fundation Period

AI Foundation

- Knowledge & AI History
- Data Exploration & Preparation

Mathematical Foundation

- Probability For Machine Learning
- Linear Algebra For Data Science
- Numerical Algorithms
- Optimization For Data Science

Development Foundation

- Introduction To Python
- Java Programming Basics (Online)
- Java & UML Programming
- Algorithm Workshop: Initiation
- Linux Administration

Probability and Statistics

for Machine Learning

Connections between probability and machine learning

Evaluation metrics for model accuracy (accuracy, precision, recall, ….)

Bayes’ rule

Estimation/Linear regression (Maximum Likelihood (ML)!)

Probabilistic Supervised Learning

Statistical Machine Learning

Cross-entropy (e.g., objective function in NN)

What else?

Naïve Bayes Classifier

$$\boxed{P(A|B) = \frac{P(B|A)\, P(A)}{P(B)}}$$

Thomas Bayes
1702 - 1761

Image source: https://en.wikipedia.org/wiki/Naive_Bayes_classifier

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LEARNING OUTCOMES

(to understand)

- what a probability is, and its properties
- independence, probability distribution (marginal, joint and conditional), mean, variance, standard deviation, covariance and correlation
- discrete and continuous distributions
- what makes the normal distribution different (CLT)
- (and be able to use) Bayes rule
- connection between Bayes rule and Naive Bayes algorithms
- maximum likelihood estimation (optional)

Content trajectory

1. Probability properties and statistical tools

What a probability is. Its properties. Random variable, independence, joint and marginal probability distributions

Statistical tools : mean, variance and standard deviation.

Covariance and correlation (briefly introduced)

2. Probability distributions

Discrete and continuous RVs.

Normal distribution and CLT

3. Conditional probabilities and Bayes theorem

Conditional probability, Bayes theorem

Relation to Naive Bayes algorithm

Maximum likelihood estimation

Note on Course Structure

- Adjustments can be made depending on the level of the students.
- If all the students are already confident with a lot of content, new topics may be introduced.
- Grading can also be adjusted.

Sessions Contents

Session 1: Basics of probability

Session 2: Independence, probability distribution and statistical tools

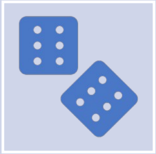
Session 3 & 4: Discrete, continuous probability distributions and the special case of the normal distribution

Session 5: Conditional probability, Bayes theorem, and Naïve Bayes Classifier.

Basic Probability Concepts

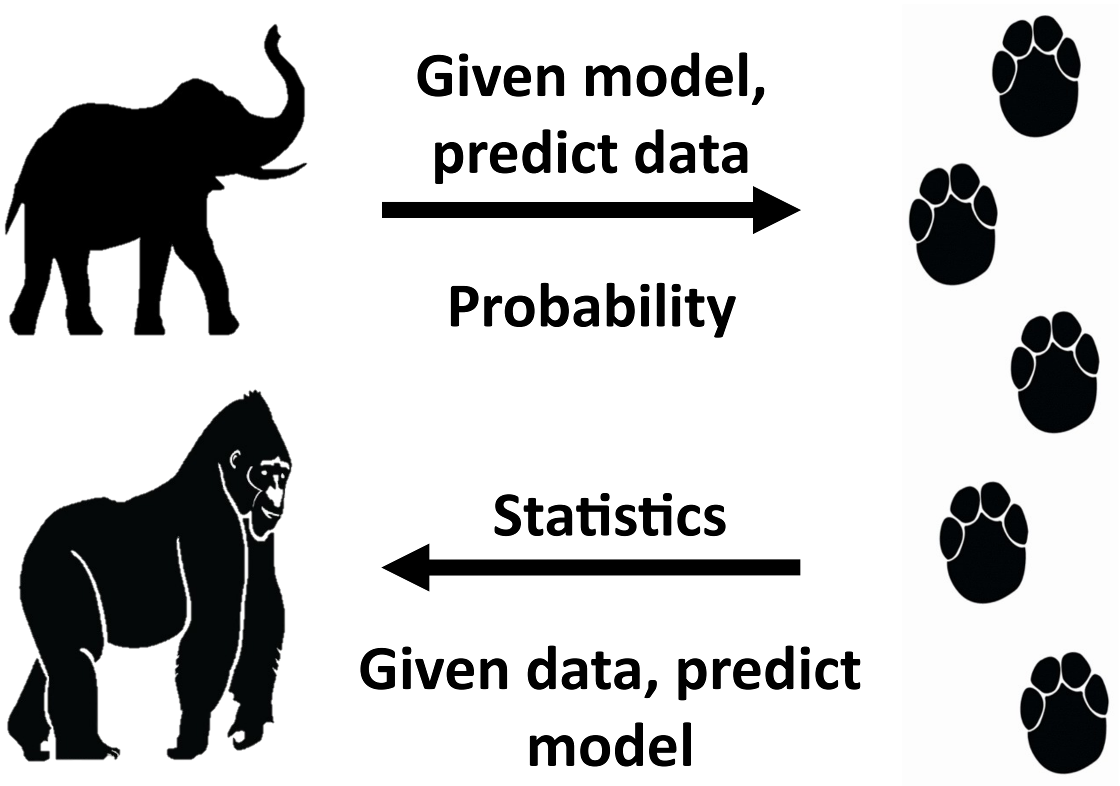
- What is a probability?
- Random experiment
- Outcomes
- Sample space
- Events
- Discrete and Continuous sample spaces

Probability vs Statistics



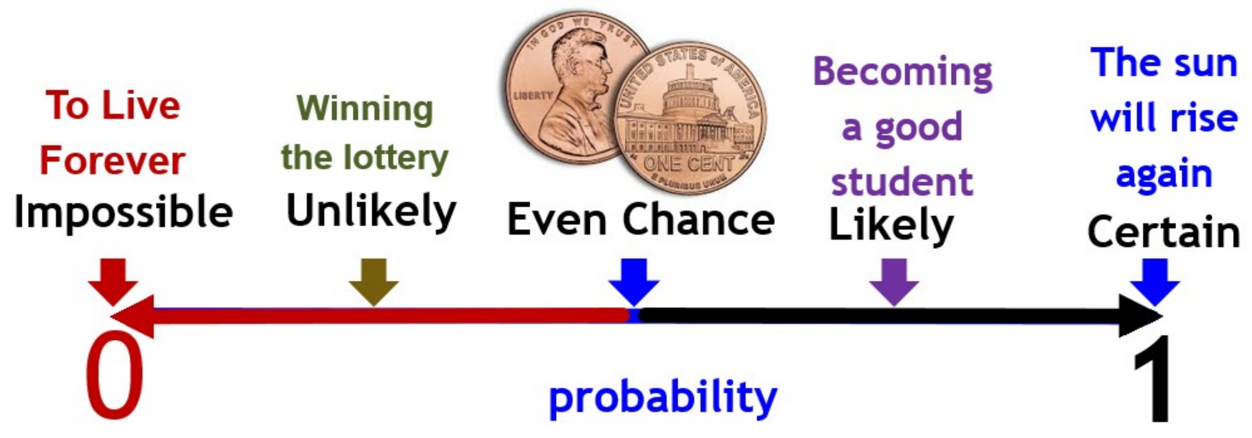
Probability deals with predicting the likelihood of future events, while

Statistics involves the analysis of the frequency of past events.



Probability

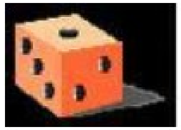
- number used to quantify the likelihood, or chance, that an outcome of a ***random experiment*** will occur.
- We can obtain probabilities from:
    - Subjective experience
    - Observations and Data collection
    - Computations using mathematical rules



Random Experiment

- an experiment that can result in different ***outcomes***, even though it is repeated the same manner every time.
- An **outcome** is the result of **a single trial**.
- **Sample space** '***S***': is the ***set*** of all possible outcomes
- **Event**: subset of **S**.

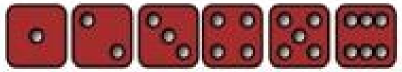
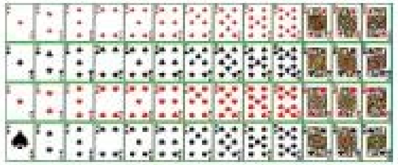
- **Flipping a coin once**



- **Rolling a die once.**

- **Pulling a card from a deck.**

Sample spaces, **outcomes**, and **events**

Experiment	Sample space
Tossing coin once	Head, Tail 
Roll a die 	1,2,3,4,5,6 
Answer a true or false question	True, False
Draw a card from a regular deck	52 cases 

Classical Probability

The (classical) probability, $P(E)$, of an event E is defined as:

$$P(E) = \frac{N(E)}{N(S)} = \frac{|E|}{|S|}$$

where $N(E)$ or $|E|$ is the number of elements in the set E.

(assuming all events are equally likely)

Empirical vs Theoretical probability

Experimental Probability is found by repeating an experiment and observing the outcomes.

$$P(\text{event}) = \frac{\text{number of times event occurs}}{\text{total number of trials}}$$

Example:

A coin is tossed 10 times:

A head is recorded 7 times and a tail 3 times.

$$P(\text{head}) = \frac{7}{10}$$

$$P(\text{tail}) = \frac{3}{10}$$

Theoretical Probability is what is expected to happen based on mathematics

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$$

Example:

A coin is tossed.

$$P(\text{head}) = \frac{1}{2}$$

$$P(\text{tail}) = \frac{1}{2}$$

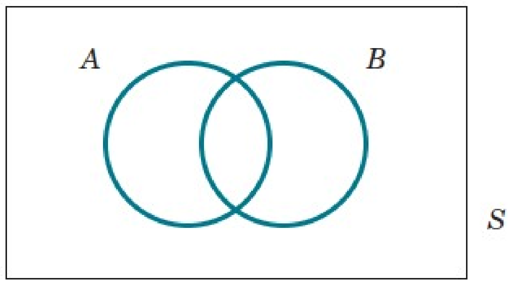
Probability as the 'Relative Frequency'

$$f(A) = \lim_{N \to \infty} \frac{N(A)}{N}$$

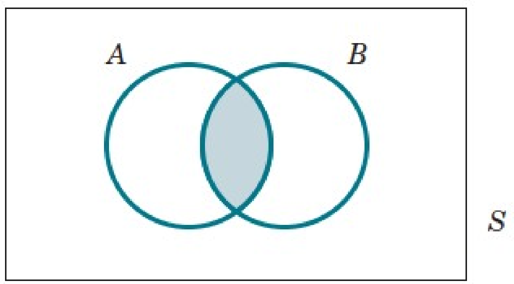
Basic Probability Concepts

Event
An **event** is a subset of the sample space of a random experiment.

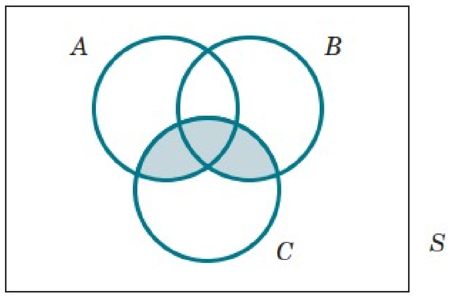
Sample space S with events A and B



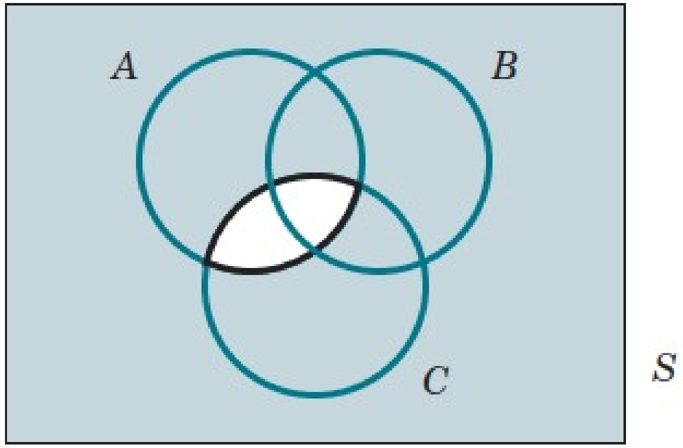
$A \cap B$



$(A \cup B) \cap C$



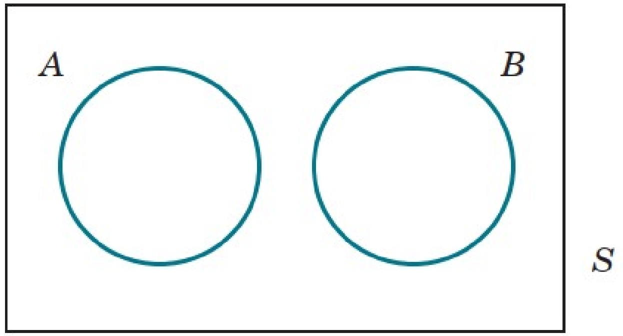
$(A \cap C)'$



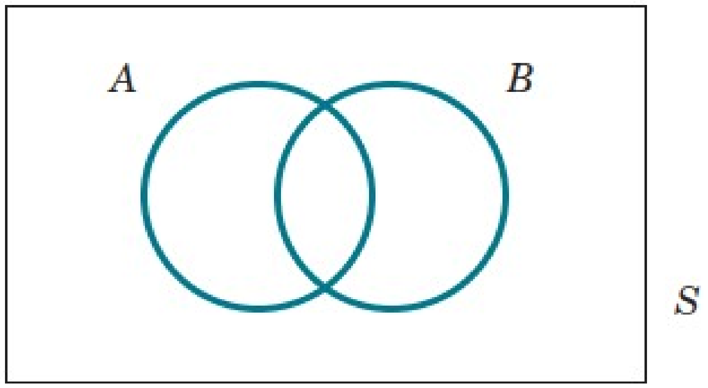
Mutually Exclusive Events
Two events, denoted as E_1 and E_2, such that

$$E_1 \cap E_2 = \varnothing$$

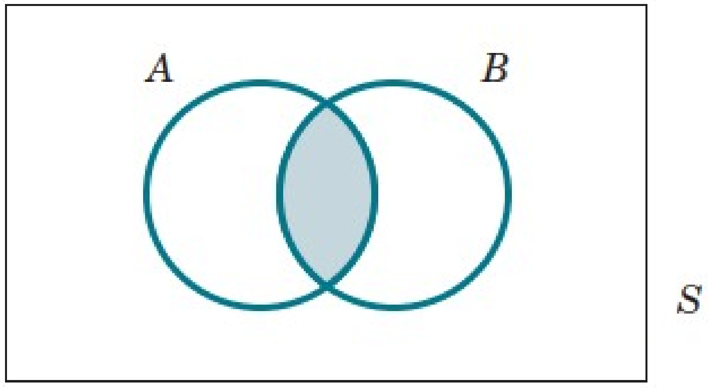
are said to be **mutually exclusive**.

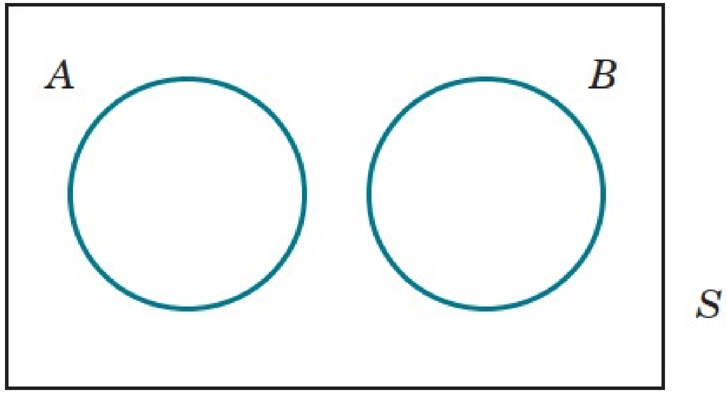


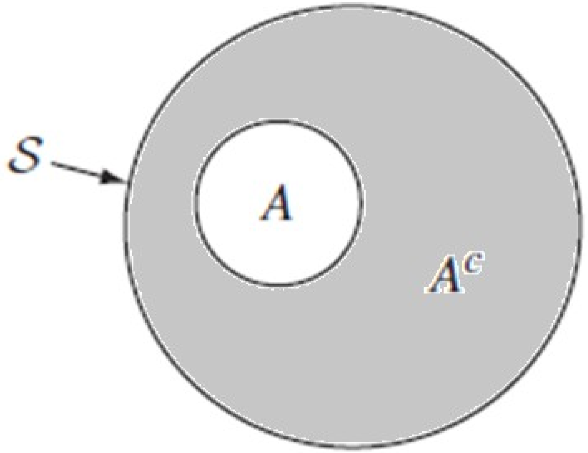
Sample space S with events A and B



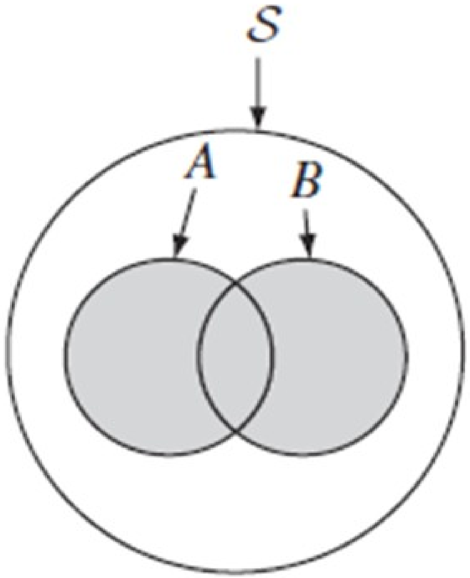
$A \cap B$



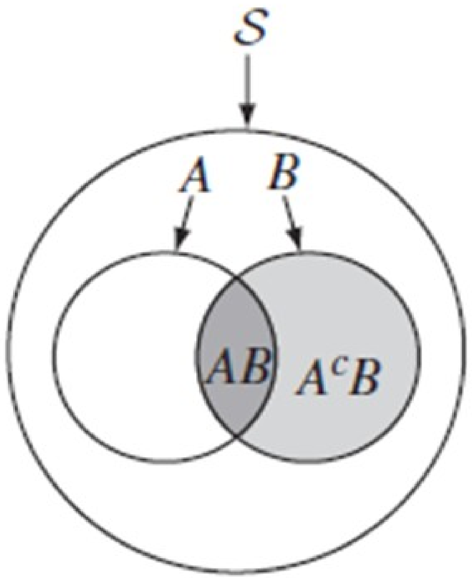




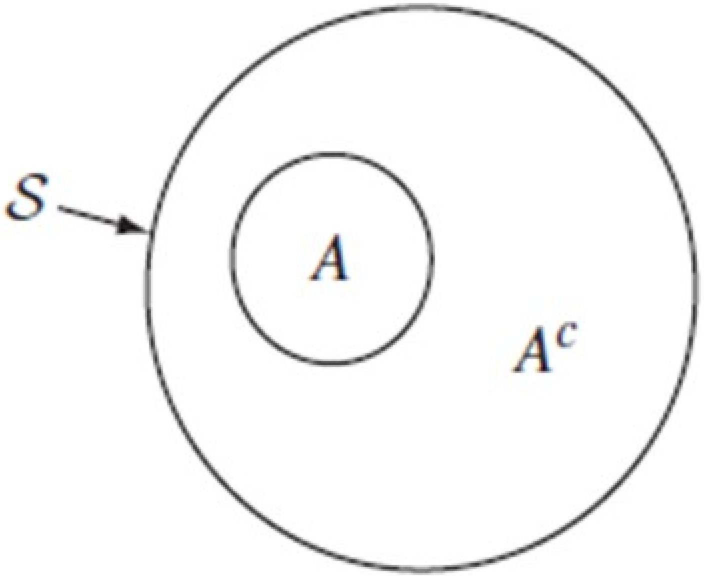
A^c



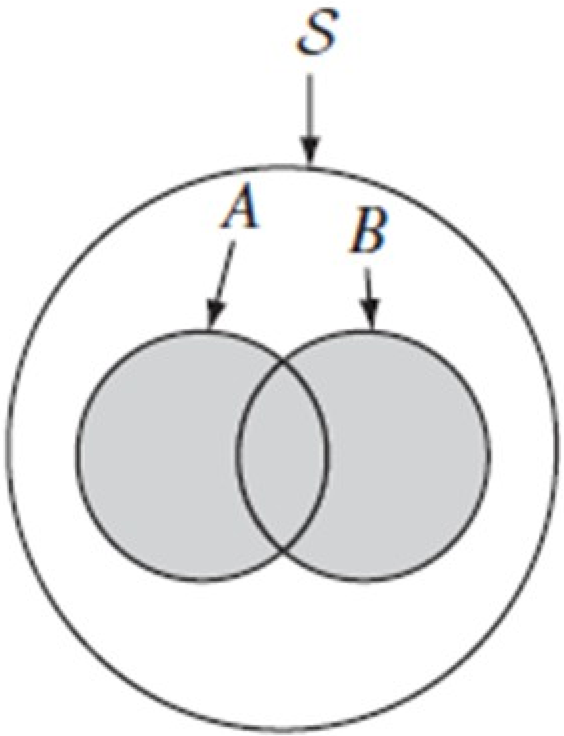
$A \cup B = B \cup A$



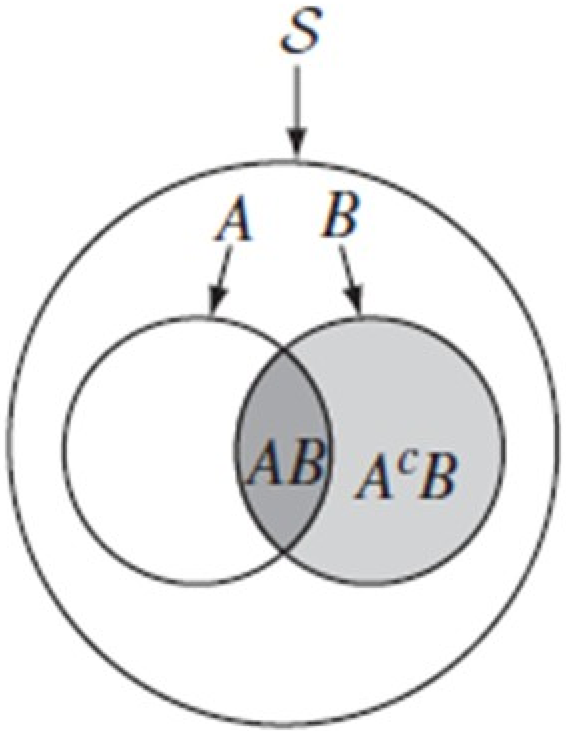
AB and A^cB



A^c



$A \cup B = B \cup A$



AB and A^cB

Fundamentals of Probability

Axioms of Probability

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties:

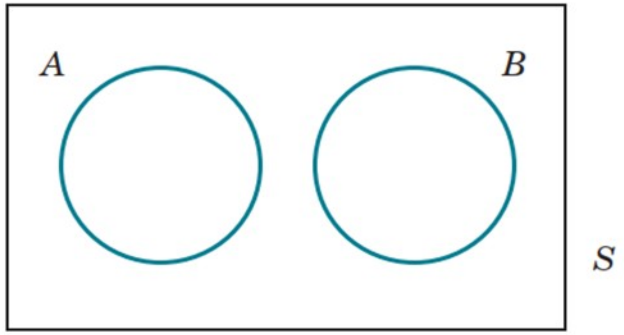
(1) $P(S) = 1$ where S is the sample space
(2) $0 \leq P(E) \leq 1$ for any event E
(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \emptyset$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

E1 and E2 are disjoint (mutually exclusive)

- Probability of impossible\Empty event: $P(\emptyset) = 0$

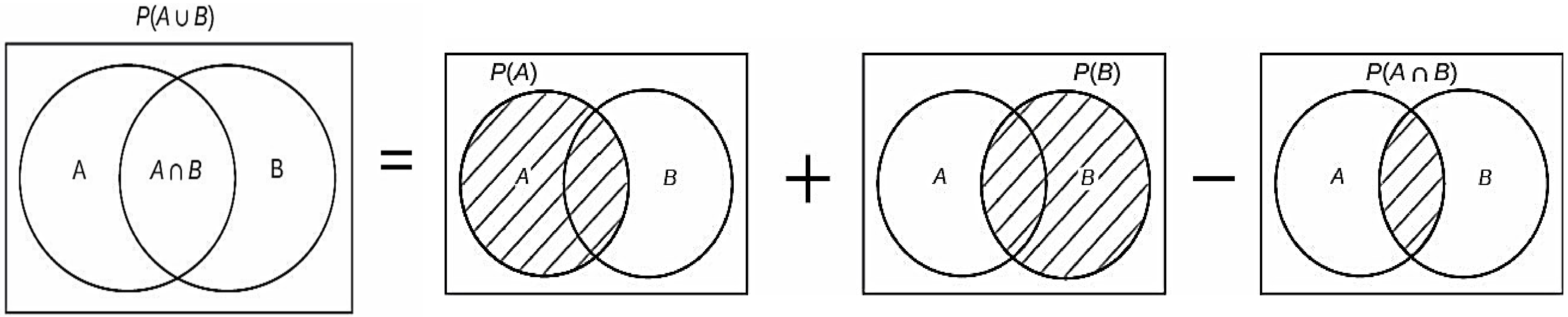
- Probability of complementary event: $P(E') = 1 - P(E)$



Union

Probability of a Union

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

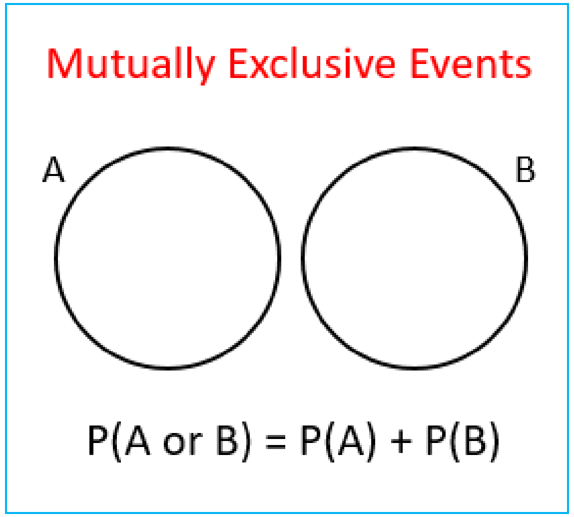


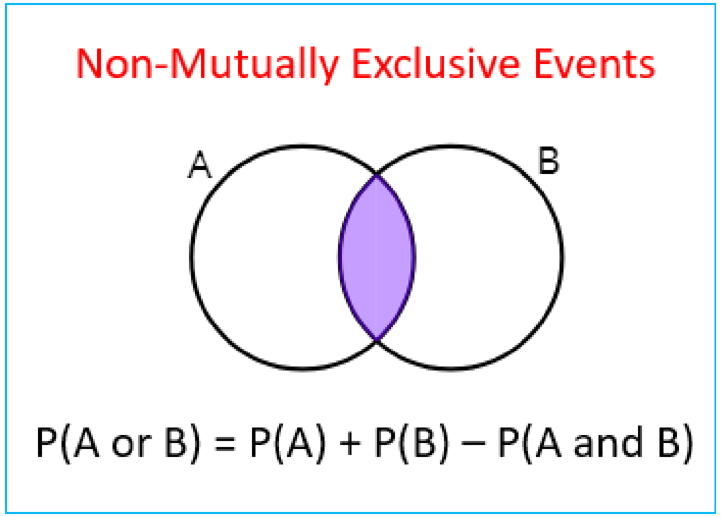
Example

Test your understanding

- S={1,2,3,4,5,6}
- A={1,3,5} B={2,4}   C={4,5,6}

- P(S) = 1
- P(A) = 3/6 = 0.5
- P(B) = 2/6 = 0.333
- P(A ∪ B) = 5/6
- P(A ∪ C) = P(A) + P(C) – P(AC)
- □ A = {1,3,5} C={4,5,6} AC={5}
- 3/6 + 3/6 – 1/6 = 5/6

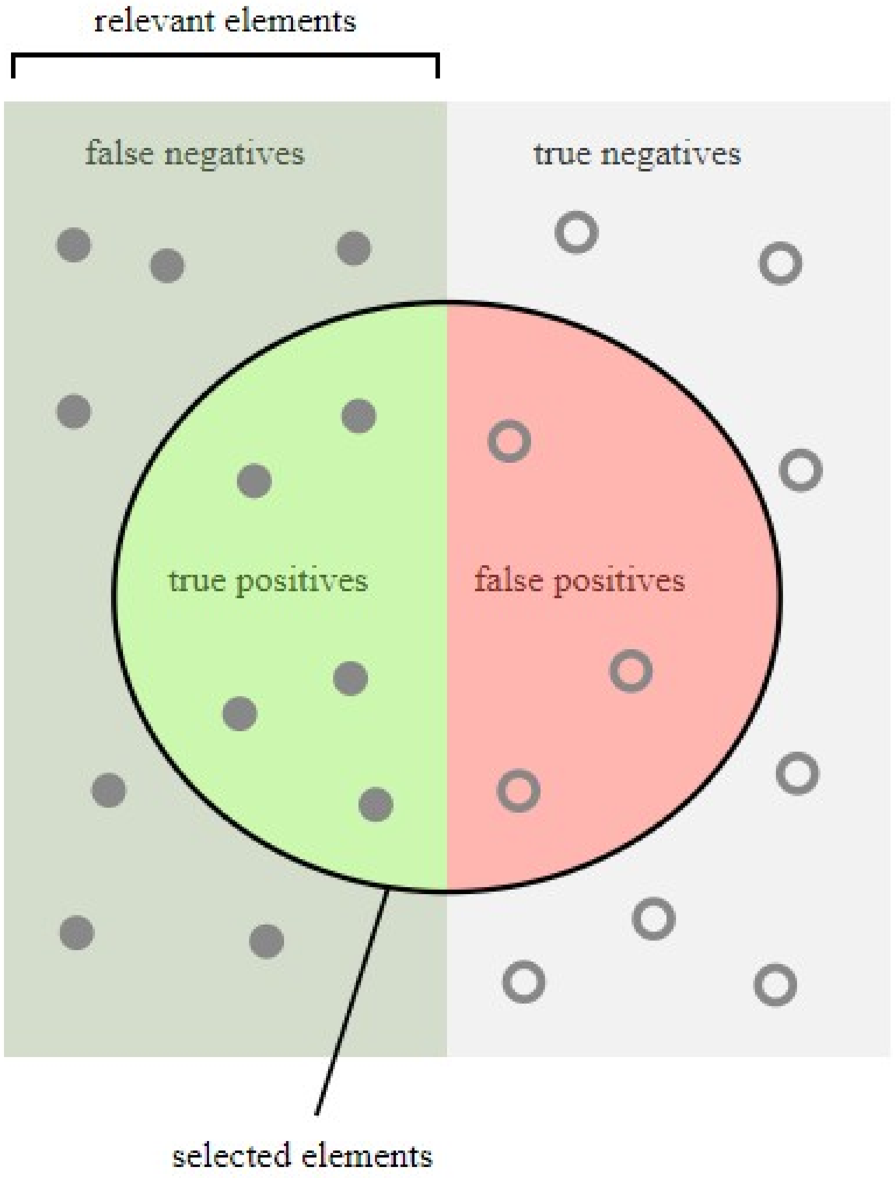




Example (performance metrics): Precision, Recall, Accuracy …

$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$



To be discussed in more detail later in the course

Independence of random events

- If the occurrence or nonoccurrence of either events A and B have no effect on the other, then A and B are independent.

(To be discussed in detail in sessions 2, 4 & 5)

If A and B are independent, then

1. $P(A|B) = P(A)$
2. $P(B|A) = P(B)$
3. $P(A \cap B) = P(A)P(B)$
4. More generally than the above, $P(\bigcap_{i=1}^{k} A_i) = \prod_{i=1}^{K} P(A_i)$

Independence vs mutual exclusiveness

(To be discussed in detail in sessions 2, 4 & 5)

- Events are *mutually exclusive* means that the occurrence of one event implies that none of the others can occur at the same time: i.e., ***P(A and B) = 0*.**

- Events are *independent* if the occurrence of one event does not affect the occurrence of another "i.e., has no effect on the probability of the occurrence of the other".

- **Definition:** Events *A* and *B* are called *Independent* if ***P(A and B) = P(A) P(B)***

- Example: When drawing a card:
  - Drawing a club and drawing a heart are mutually exclusive events.
  - Drawing a club and drawing a king are not mutually exclusive events

Independence of random events

Independent Events

The outcome of one event does not affect the outcome of the other.

If A and B are independent events then the probability of both occurring is

$$P(A \text{ and } B) = P(A) \times P(B)$$

Dependent Events

The outcome of one event affects the outcome of the other.

If A and B are dependent events then the probability of both occurring is

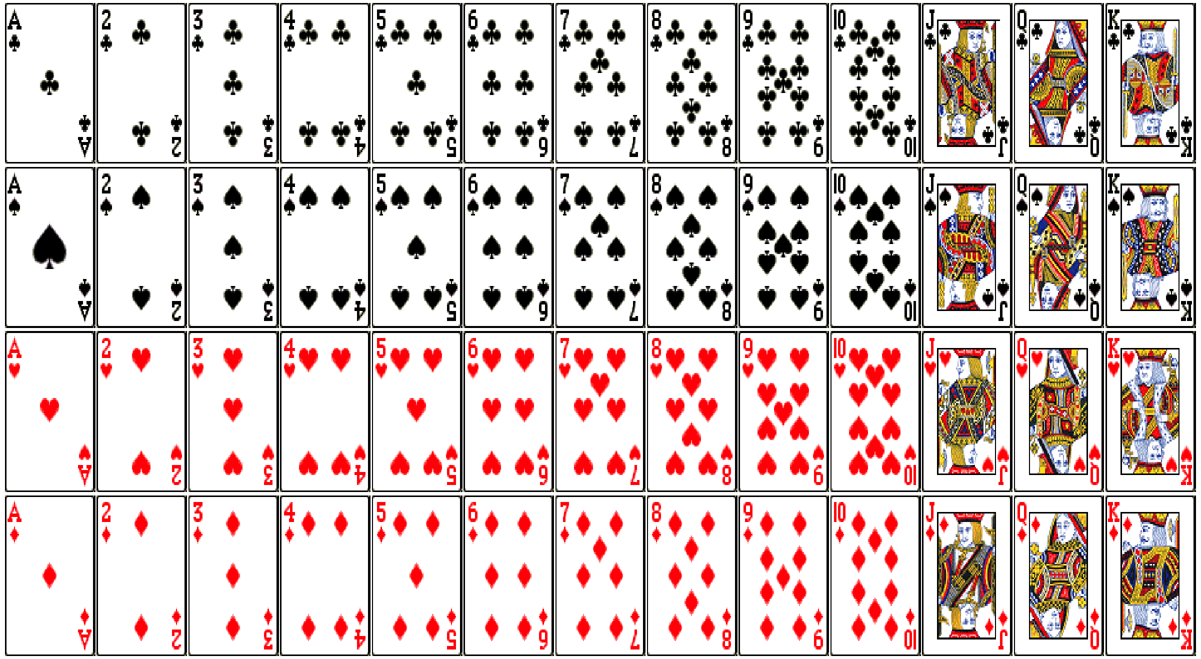
$$P(A \text{ and } B) = P(A) \times P(B|A)$$

Probability of B given A

(To be discussed in detail in session 4)

Example

A card is drawn from a standard deck. (with vs. without replacement!)



Examples and Further Discussion

Example

Test your understanding

- S = {1,2,3,4,5,6}
- A: Even number
- B: Odd Number
- C = {2,3,5}: Prime Number
- D: Number > 2
- Are A, C independent?
- P(A|C) = 1/3
- P(A) = 3/6

If A and B are independent, then

1. $P(A|B) = P(A)$
2. $P(B|A) = P(B)$
3. $P(A \cap B) = P(A)P(B)$
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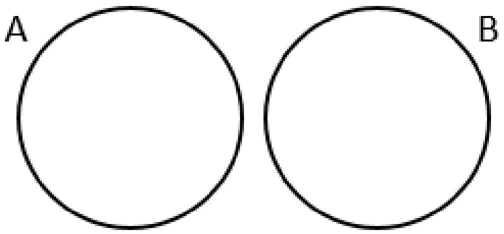
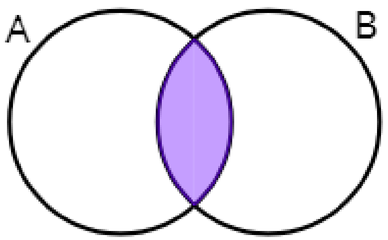
Mutually Exclusive Events



P(A or B) = P(A) + P(B)

Non-Mutually Exclusive Events



P(A or B) = P(A) + P(B) – P(A and B)

Example

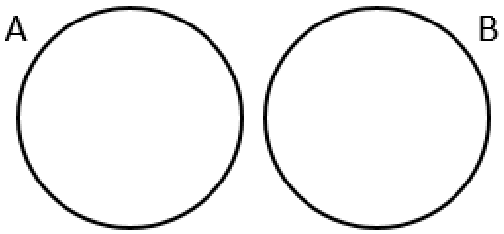
Test your understanding

- S = {1,2,3,4,5,6}
- A: {2,4,6}: Even number
- C = {2,3,5}: Prime Number
- Are A, C independent? No!
- P(A|C) = 1/3
- P(A) = 3/6 = 0.5

If A and B are independent, then

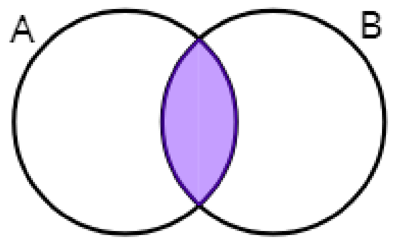
1. $P(A|B) = P(A)$
2. $P(B|A) = P(B)$
3. $P(A \cap B) = P(A)P(B)$
4. More generally than the above, $P(\bigcap_{i=1}^{k} A_i) = \prod_{i=1}^{K} P(A_i)$

Mutually Exclusive Events



P(A or B) = P(A) + P(B)

Non-Mutually Exclusive Events



P(A or B) = P(A) + P(B) – P(A and B)

Example: Tossing two dice one time

Example: Two dice are tossed.

The Random Variable is **X** = "The sum of the scores on the two dice".

Let's make a table of all possible values:

		1st Die					
		1	**2**	**3**	**4**	**5**	**6**
2nd Die	**1**	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Probability

Example: Tossing two dice one time

There are **6 × 6 = 36** possible outcomes, and the Sample Space (which is the sum of the scores on the two dice) is {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

Let's count how often each value occurs, and work out the probabilities:

- 2 occurs just once, so P(X = 2) = 1/36
- 3 occurs twice, so P(X = 3) = 2/36 = 1/18
- 4 occurs three times, so P(X = 4) = 3/36 = 1/12
- 5 occurs four times, so P(X = 5) = 4/36 = 1/9
- 6 occurs five times, so P(X = 6) = 5/36
- 7 occurs six times, so P(X = 7) = 6/36 = 1/6
- 8 occurs five times, so P(X = 8) = 5/36
- 9 occurs four times, so P(X = 9) = 4/36 = 1/9
- 10 occurs three times, so P(X = 10) = 3/36 = 1/12
- 11 occurs twice, so P(X = 11) = 2/36 = 1/18
- 12 occurs just once, so P(X = 12) = 1/36

Value of x_i	Dice Outcomes	$P_X(x_i)$
2	(1, 1)	1/36
3	(1, 2), (2, 1)	2/36 = 1/18
4	(1, 3), (2, 2), (3, 1)	3/36 = 1/12
5	(1, 4), (2, 3), (3, 2), (4, 1)	4/36 = 1/9
6	(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)	5/36
7	(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)	6/36 =1/6
8	(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)	5/36
9	(3, 6), (4, 5), (5, 4), (6, 3)	4/36 = 1/9
10	(4, 6), (5, 5), (6, 4)	3/36 = 1/12
11	(5, 6), (6, 5)	2/36 = 1/18
12	(6, 6)	1/36

Probability

- **Range of Values**
- We could also calculate the probability that a Random Variable takes on a range of values.

Example (continued) What is the probability that the sum of the scores is 5, 6, 7 or 8?

In other words: What is **P(5 ≤ X ≤ 8)**?

$$
\begin{aligned}
P(5 \le X \le 8) &= P(X=5) + P(X=6) + P(X=7) + P(X=8) \\
&= (4+5+6+5)/36 \\
&= 20/36 \\
&= 5/9
\end{aligned}
$$

Example

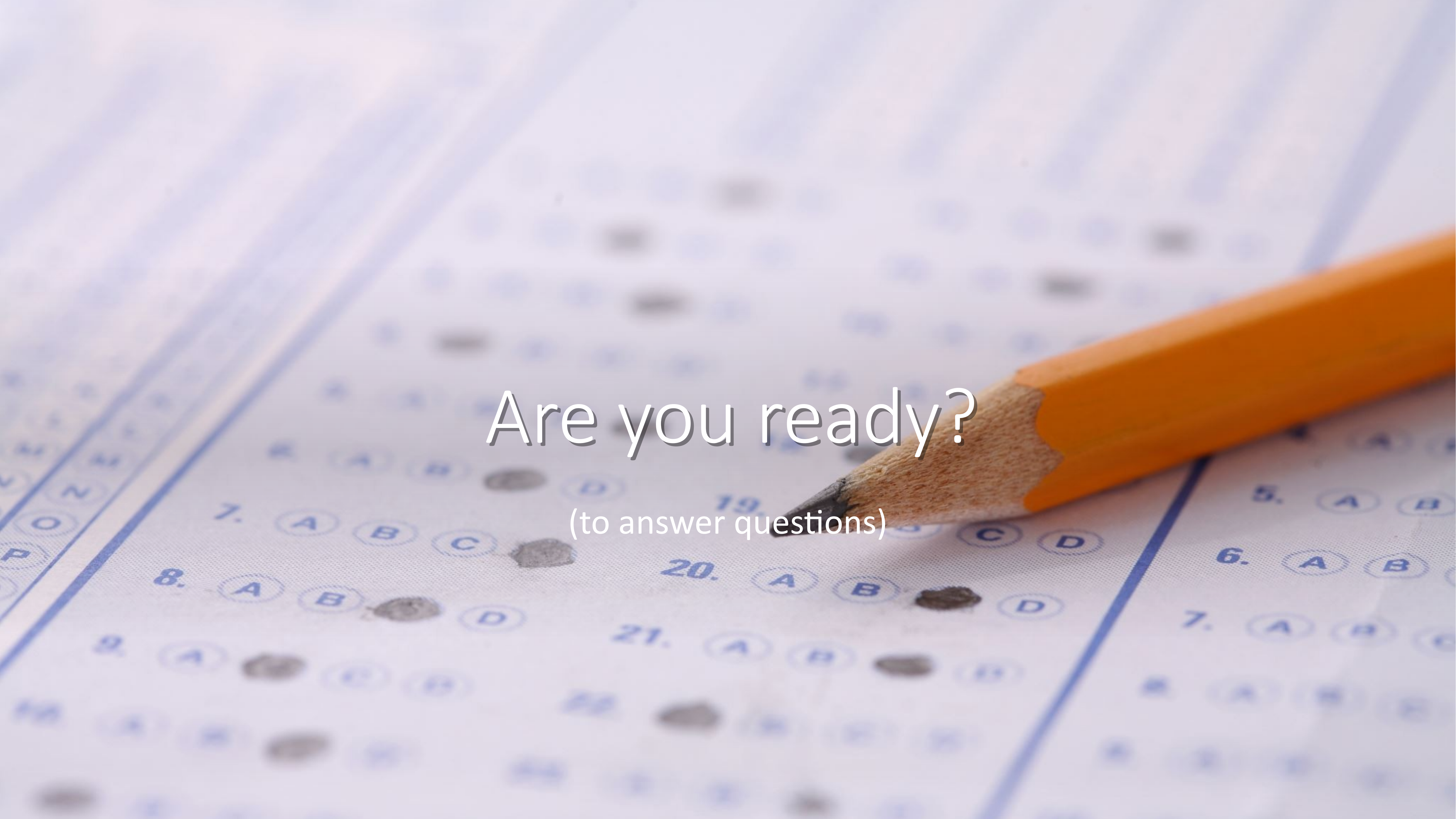
Coin 1	Coin 2	Coin 3	Prob
H	H	H	1/8
H	H	T	1/8
H	T	H	1/8
H	T	T	1/8
T	H	H	1/8
T	H	T	1/8
T	T	H	1/8
T	T	T	1/8

Test your understanding.

- If you flipped a coin 1 time, find the probability of getting **a head**. P(H) = 1/2
- If you flipped a coin 2 times, find the probability of getting **two heads**. P(H1 H2) = 1/4 = P(H1)xP(H2) = .5 x .5 = 1/4
- If you flipped a coin 3 times (or flipping three coins), find the probability of getting **two heads** and **one tail**.

P(2 Heads and 1 Tail) = 3xP(H1H2T3) = 3*1/8 = 3/8

Questions for the instructor?



Are you ready?

(to answer questions)

Textbooks

- Schaum's outline, "Probability, Random variables, and Stochastic Process".
- Papoulis and Pillai, "Probability, Random Variables and Stochastic Process".

Python-based Practice

- Kindly check these two notebooks
    1. probability.ipynb
    2. simulating_coin_flips.ipynb